

Exercise

Build a job listing site using Next.js. Aim to spend about two to three hours.

Important: We care more about how you work than completeness. Ensure that you regularly commit your work at reasonable milestones as we like to assess how you approach a problem. A submission lacking a detailed commit history is unlikely to progress. You do not need to complete every requirement to progress; depth over breadth is preferred.

Background

This exercise reflects the real work you would be doing from day one. Our website is a Next.js application that pulls job listings from an external feed, renders them with full technical SEO, and needs to convert visitors into applicants. Your task covers that full problem: routing, SEO, analytics, and performance.

Job Feed

Use the following fixture as your data source. Serve it however is convenient: a Next.js API route, a local JSON file read at build time, or any other approach. The important constraint is that your data fetching must be cleanly separated from your rendering layer.

The feed intentionally includes jobs with a null closingDate. Your UI should handle this gracefully throughout.

```
[
  {
    "id": "JOB-001",
    "title": "Paediatrician",
    "slug": "paediatrician",
    "location": "Auckland, NZ",
    "type": "Full-time",
    "department": "Paediatrics",
    "postedDate": "2026-04-01",
    "closingDate": "2026-05-15",
    "salary": { "min": 220000, "max": 280000, "currency": "NZD" },
    "description": "<p>We are seeking a Consultant Paediatrician to join our multidisciplinary team delivering specialist child health services across the Auckland region. You will provide expert assessment, diagnosis, and management for a broad range of paediatric conditions, and contribute to clinical governance and teaching.</p>",
    "requirements": [
      "Fellowship of the Royal Australasian College of Physicians (FRACP) or equivalent",
      "Vocational registration with the Medical Council of New Zealand",
      "Demonstrated experience in general paediatrics or a paediatric subspecialty"
    ]
  },
  {
```

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    "id": "JOB-002",
    "title": "Emergency Medicine Physician",
    "slug": "emergency-medicine-physician",
    "location": "Wellington, NZ",
    "type": "Full-time",
    "department": "Emergency Medicine",
    "postedDate": "2026-04-10",
    "closingDate": null,
    "salary": { "min": 230000, "max": 290000, "currency": "NZD" },
    "description": "<p>Join our busy emergency department as a Consultant
Emergency Medicine Physician. You will manage undifferentiated and acutely
unwell patients across all age groups, lead resuscitation efforts, and
support junior medical staff in a high-volume tertiary setting.</p>",
    "requirements": [
        "Fellowship of Australasian College for Emergency Medicine (FACEM)
or equivalent",
        "Current vocational registration with the Medical Council of New
Zealand",
        "Experience in a tertiary or mixed-level emergency department"
    ]
},
{
    "id": "JOB-003",
    "title": "General Practitioner",
    "slug": "general-practitioner",
    "location": "Queenstown, NZ",
    "type": "Full-time",
    "department": "Primary Care",
    "postedDate": "2026-04-15",
    "closingDate": "2026-06-01",
    "salary": { "min": 180000, "max": 240000, "currency": "NZD" },
    "description": "<p>We are looking for a vocationally registered
General Practitioner to join a well-established primary care practice in
Queenstown. You will provide comprehensive, patient-centred care to a
diverse community and have opportunities to develop special clinical
interests.</p>",
    "requirements": [
        "Vocational registration as a General Practitioner with the Medical
Council of New Zealand",
        "Fellowship of the Royal New Zealand College of General
Practitioners (FRNZCGP) or equivalent",
        "Strong commitment to continuity of care and community health"
    ]
},
{
    "id": "JOB-004",
    "title": "Consultant Anaesthetist",
    "slug": "consultant-anaesthetist",
    "location": "Christchurch, NZ",
    "type": "Full-time",
    "department": "Anaesthesia",
    "postedDate": "2026-04-20",
    "closingDate": "2026-05-30",
    "salary": { "min": 250000, "max": 320000, "currency": "NZD" },
    "description": "<p>An exciting opportunity exists for a Consultant
Anaesthetist to join our perioperative team in Christchurch. You will

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provide anaesthetic services across a range of surgical specialties,
participate in acute pain and obstetric anaesthesia, and contribute to
departmental quality improvement.</p>",
  "requirements": [
    "Fellowship of the Australian and New Zealand College of
    Anaesthetists (FANZCA) or equivalent",
    "Vocational registration with the Medical Council of New Zealand",
    "Broad anaesthetic experience including obstetrics and acute pain
    management"
  ]
},
{
  "id": "JOB-005",
  "title": "Psychiatrist",
  "slug": "psychiatrist",
  "location": "Remote (NZ)",
  "type": "Contract",
  "department": "Mental Health",
  "postedDate": "2026-04-22",
  "closingDate": null,
  "salary": { "min": 210000, "max": 270000, "currency": "NZD" },
  "description": "<p>We are recruiting a Consultant Psychiatrist to
  provide telehealth psychiatric services to patients across rural and
  regional New Zealand. You will conduct assessments, manage complex cases,
  and work collaboratively with community mental health teams and primary
  care providers.</p>",
  "requirements": [
    "Fellowship of the Royal Australian and New Zealand College of
    Psychiatrists (FRANZCP) or equivalent",
    "Vocational registration with the Medical Council of New Zealand",
    "Experience with telehealth delivery and working with rural or
    underserved populations"
  ]
}
]

```

Task

1. Job Listings Index

- Display all jobs with title, location, type, department, and posted date
- Support filtering by department and employment type; filters must be reflected in the URL as query parameters so filtered views are shareable and bookmarkable
- Canonical tags on filtered views must point to the unfiltered index page

2. Job Detail Page

- Render full job details at /jobs/[slug] using Next.js dynamic routing
- Include title, location, type, department, salary range, posted date, closing date (or "Open until filled" if null), description, and requirements

- An Apply button that fires an analytics tracking event on click; you may stub this as a `window.dataLayer` push, but the structure should reflect how a real GTM implementation would work
- Return a 404 with appropriate metadata for any slug that does not exist in the feed

3. SEO

- Use the Next.js Metadata API on every page: title, meta description, Open Graph tags, and a canonical tag
- Job detail pages must include a JobPosting JSON-LD structured data block with all required and relevant recommended fields
- Generate a `sitemap.xml` that includes all job listing URLs and the index page

4. Technical

- TypeScript throughout with well-defined types for the job data model
- The data fetching layer must be fully separated from the rendering layer
- Write at least one unit test and one integration test for a page route
- Deploy to Vercel or another platform (optional, we can always run it locally)

What to Submit

- Public repository link containing:
 - Solution code
 - Commit history that shows your process
 - README that includes:
 - Clear setup, run, and test steps
 - If deployed, the live URL
 - How a real external job feed would be plugged in to replace the fixture
 - Your rendering strategy choice (SSG / ISR / SSR) and the reasoning behind it
 - How your analytics stub would connect to a real GTM or analytics implementation
 - A section reflecting on trade-offs you made during the exercise
 - A section describing next steps you would take if you had more time
 - Your view on what is still vital but missing before this could be considered production-ready

Rules on AI Usage

You may use AI tools to assist with generating code or other parts of the solution. The final submission must be your own work and you must be able to understand, explain, and justify all parts of your submission during the technical interview, regardless of whether it was AI-assisted. This is the same standard we follow internally at Medenterprises.

In your README, clearly note if any part of the work was generated or significantly influenced by AI.

Assessment Criteria

We will look for:

- **A detailed commit history, with regular commits (most important!)**
- Data layer design: is the fetching layer genuinely decoupled so any source could be swapped in?
- TypeScript quality: accurate and safe types for the job data model
- Routing and dynamic page generation: correct use of Next.js dynamic routes and URL-reflected filter state
- SEO depth: Metadata API usage, canonical tags on filtered views, correct and complete JobPosting JSON-LD, and sitemap
- Analytics stub: structure that reflects a real GTM implementation, not just a placeholder comment
- 404 handling: correct status code and metadata for unknown slugs
- UI quality: clean, responsive layout; graceful handling of null closing dates throughout
- Test quality: meaningful tests, not just happy-path smoke tests
- Clear documentation with thoughtful reflection on limitations and production readiness